



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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Institute of Technology
(MJIT)

SECJ3104: Applications Development

Final Report

PROJECT Lab and Equipment Booking System

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1.0 INTRODUCTION

At Universiti Teknologi Malaysia Kuala Lumpur, laboratory facilities and specialized equipment are managed by PPMU (Pusat Pengurusan Makmal Universiti). The current booking process relies on Google Forms, email communication, and manual staff verification. This causes several issues, such as slow request handling, limited booking visibility, difficulty checking timetable conflicts, and the lack of a centralized approval and record-tracking system.

To solve these problems, the **Lab and Equipment Booking System** was developed as a web-based platform for managing laboratory resource bookings. The system allows students and staff to browse labs and equipment, check schedules, submit booking requests, and track their request status in one place. It also helps reduce booking conflicts by replacing scattered manual communication with a more structured workflow.

Before a request is submitted, the system checks important booking rules such as office hours, advance booking requirements, booking date limits, weekend restrictions, equipment availability, class timetable clashes, and conflicts with existing bookings. Users can also add multiple equipment items into one request, provide requester and lecturer details, and download booking-related documents such as receipts and quotations.

The system includes a role-based approval process. PIC users manage booking access tokens, Unit Leaders review and recommend requests, and PPMU users make the final approval decision, update equipment availability, manage pricing, and confirm request details. Email notifications are also used to keep users and staff informed about important booking updates.

Overall, the Lab and Equipment Booking System improves the manual booking process by making it more organized, transparent, and easier to manage. This report presents the system's design, implementation, and testing to show how it supports laboratory and equipment booking at UTM Kuala Lumpur.

2.0 USER STORY

Student / Staff

US001 - Account Access

As a Student/Staff, I want to register an account, log in, and reset my password so that I can securely access the Lab and Equipment Booking System.

Acceptance Criteria:

- The system allows Student/Staff to register an account using valid required details.
- The system prevents registration if the email has already been used.
- The system allows registered users to log in using email and password.
- The system rejects login attempts with incorrect credentials.
- The system allows users to request and complete password reset.
- The system allows users to log in using the new password after reset.
- The system redirects users to the correct interface based on their role.

US002 - Browse Labs, Equipment, and Schedule

As a Student/Staff, I want to browse labs and equipment so that I can view available resources before making a booking request.

Acceptance Criteria:

- User can search labs or equipment by keyword.
- User can filter by programme.
- User can open a lab and view its equipment list, availability status, price, staff details, and weekly schedule.
- Unavailable or maintenance equipment cannot be selected for booking.

US003 - Create Booking Request

As a Student/Staff, I want to add the required laboratory/equipment to my request after browsing, so that submit my reservation.

Acceptance Criteria:

- Selected items are displayed in the Complete Booking page.
- Users must fill in required information, such as User details and PIC details.
- User must enter a valid 6-character PIC Token.

- The booking application can only be submitted after all required fields have been filled.
- After submission, the system will send an email to the user informing them that the order has been accepted and is awaiting approval.

US004 - View and Cancel Booking Records

As a Student/Staff, I want to view my submitted bookings and cancel pending requests so that I can view my booking history.

Acceptance Criteria:

- User can view only their own booking records.
- User can filter booking records by status.
- User can open booking details and review submitted information.
- User can cancel requests that are still pending unit leader or PPMU processing.
- Processed, rejected, or already cancelled requests cannot be cancelled again.

US005 - Download Quotation and Receipt

As a Student/Staff, I want to download the booking receipt and quotation so that I can keep a copy of my booking and payment details.

Acceptance Criteria:

- User can download a receipt after successfully submitting a booking request.
- Receipt includes booking details such as user information, selected lab/equipment, date, time, and request status.
- User can download quotation after the booking has been processed or approved by PPMU.
- Quotation includes pricing details, booking item details, and total amount.
- Downloaded receipt and quotation are generated in PDF format.

PIC

US006 - Generate and Assign PIC Token

As a PIC, I want to generate and assign a token to a user so that the user can verify permission before booking lab.

Acceptance Criteria:

- Only PIC users can access the Token Generation page.
- PIC can search for a registered user by email.
- PIC can generate a 6-character token and assign it to the selected user.
- Assigned token shows an expiry date and cannot be assigned twice in the same action.

US007 - View Assigned Token

As a PIC, I need to view the assigned PIC tokens so that I can easily manage the tokens that have been generated.

Acceptance Criteria:

- PIC allows you to view a list of generated or assigned tokens.
- Each token displays its generation date, expiration date, token code, assigning user, and status.
- PIC can identify whether a token is active or expired.

Unit Leader

US008 - Unit Leader Recommendation

As a Unit Leader, I want to review booking requests and recommend or reject each request so that only suitable requests proceed to PPMU.

Acceptance Criteria

- Only unit leader users can access the Unit Leader dashboard.
- Dashboard shows totals, pending, recommended, and rejected counts.
- Unit leader can search, filter, sort, and open request details.
- Unit leader can recommend or reject individual items or apply a bulk decision with remarks.
- Recommended items move to PPMU review; rejected items do not proceed.

PPMU

US009 - PPMU Final Approval

As a PPMU, I want to make the final decision on unit-leader-recommended items so that approved bookings can be processed and quotation documents can be issued.

Acceptance Criteria

- Only PPMU users can access the PPMU dashboard.
- PPMU dashboard lists only items eligible for final review.
- PPMU can approve or reject individual items or bulk review items with remarks.
- Only unit-leader-recommended items can be reviewed by PPMU.
- Processed approved bookings generate quotation information and notification emails where configured.

US010 - Manage Equipment Availability

As a PPMU, I want to update equipment price and availability so that booking users see accurate catalog information.

Acceptance Criteria

- Only PPMU users can access Edit Equipment.
- PPMU can search labs and equipment.
- PPMU can update equipment price per hour.
- PPMU can set equipment status to Available, Under Maintenance, or

Unavailable.

- Booking page immediately reflects updated price and availability status.

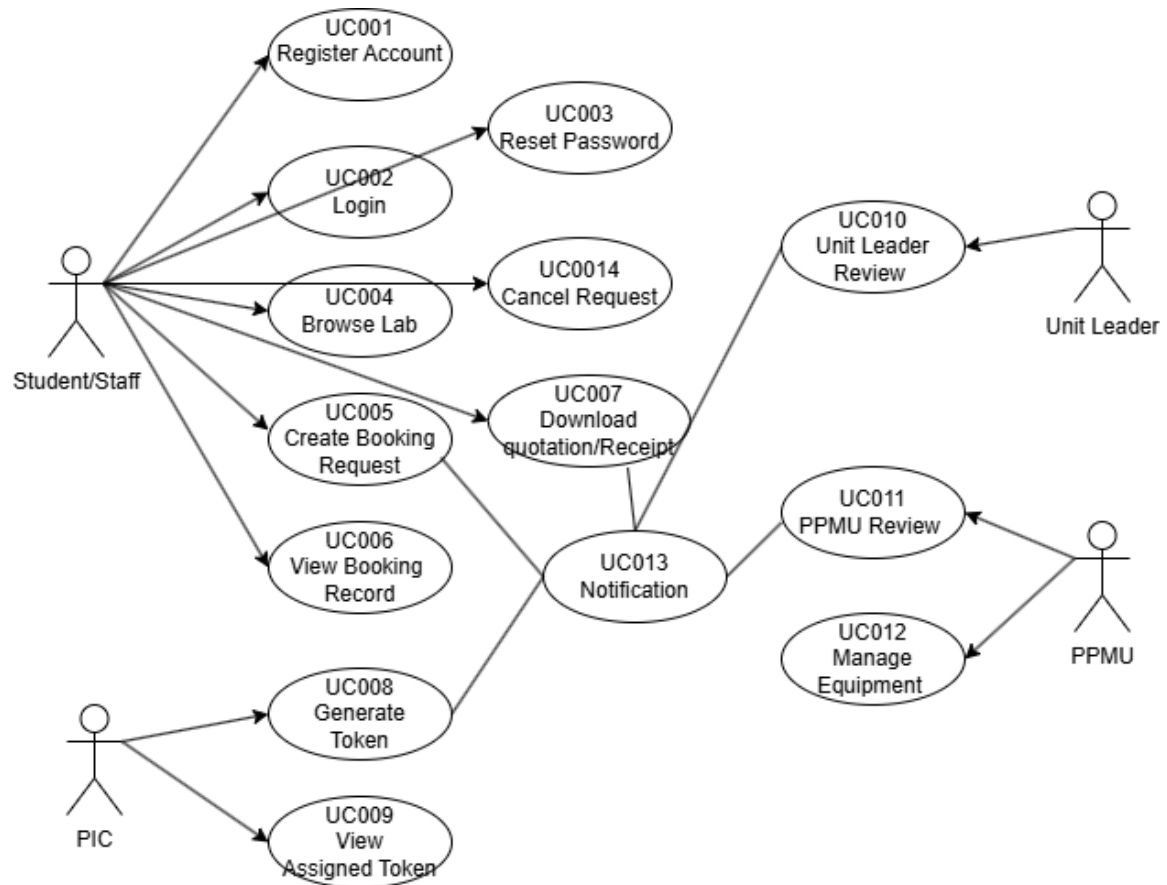
US011 – Notification

As a user, I want to receive notifications about my booking request so that I can know the latest status of my reservation.

Acceptance Criteria

- User receives an email after successfully submitting a booking request.
- User receives a notification when the booking request is approved or rejected.
- Notification includes important booking details such as request status, lab/equipment, date, and time.
- Users can receive tokens allocated by PIC.

3.0 USER CASE DIAGRAM



UC ID	UC Description
UC001	Register a new user account using valid university details.
UC002	Log in to the system using registered email and password.
UC003	Reset or update account password when needed.
UC004	Browse labs and equipment with details, availability, and schedule information.
UC005	Create and submit a lab/equipment booking request.
UC006	View submitted booking records and request status.
UC007	Download booking receipt or quotation PDF.
UC008	Generate a PIC token to authorize a user for booking.
UC009	View and manage assigned PIC tokens.
UC010	Review booking requests and provide Unit Leader recommendation.
UC011	Perform PPMU final review and approve or reject booking requests.
UC012	Manage equipment price, details, and availability status.
UC013	Send and receive booking-related notifications.
UC014	Cancel an eligible booking request.

4.0 UI DESIGN

To enhance the clarity of designing and prototyping navigation, we have assigned numbers and names to each screen. Below is the numbered flow of screens and their corresponding names.

1. **Register Page:** Allows new users to create an account by entering their username, university email, password, and role.
2. **Login Page :** Allows registered users to log in using their email and password and access features based on their role.
3. **Reset Password Page:** Allows users to reset or update their password when they forget it or need to change it.
4. **Browse Lab Page:** Displays available labs and equipment with details such as location, staff contact, equipment status, price, and schedule availability.
5. **Booking Request Page:** Allows Student/Staff to select equipment, enter booking details, provide requester and lecturer information, and submit a booking request.
6. **Booking Record Page:** Shows submitted booking requests, including request status, equipment details, booking time, price, and available actions such as cancel or download documents.
7. **Generate Token Page:** Allows PIC users to generate the required Token and assign it to the user.
8. **View Assigned Token Page:** Allows PIC users to view, sort, and expire subscription tokens assigned to them.
9. **Unit Leader Review Page:** Allows Unit Leaders to review booking requests, recommend or reject items, and add remarks.
10. **PPMU Review Page:** Allows PPMU users to make the final approval decision, edit final prices, approve or reject items, and add remarks.
11. **Management Equipment Page:** Allows PPMU users to update equipment price and availability status.

1.Register Page:

Lab Booking System

Create a new account

Username

Email

Use your @graduate.utm.my email address.

Password

Role

Student

PIC accounts are assigned by the system administrator after your account is created.

Register

Already have an account? [Login](#)

2. Login Page

Lab Booking System

Sign in with your university account

Email

Use your @graduate.utm.my email address.

Password

[Forgot password?](#)

Login

Don't have an account? [Register](#)

3. Reset Password Page

Reset Password

Enter your new password below

New Password

Confirm New Password

Update Password

4. Browse Lab Page

Booking

Browse labs, choose equipment, and add each item to one request.

Booking Instructions

1. Application shall be made at least 3 working days before usage.
2. All information provided in this form must be TRUE upon submission.
3. If the equipment requested from applicants are to be brought outside the laboratory / workshop, an application letter endorsed by Supervisor / Project Leader / Lecturer needs to be submitted to the Director of Administration of Deputy Vice-Chancellor (Research & Innovation), UTMKL.
4. The office has the right to reject any activity from the applicant if the activities are suspected to have high risks to the staff / environment and/or can cause damage to the instrument.
5. For further inquiries on the availability of laboratory and equipment, kindly contact the staff of the respective laboratory.

Q Search labs or equipment...

All Programmes

Showing 19 labs



40 equipment available

BENGKEL AUTOMOTIF

M105

Automotive workshop for vehicle systems, diagnostics, and maintenance training.

AUTOMOTIVE ENGINEERING

40 / 42 equipment available

Select



1 equipment available

BENGKEL AUTOMOTIF

M106

Automotive workshop for vehicle systems, diagnostics, and maintenance training.

AUTOMOTIVE ENGINEERING

1 / 1 equipment available

Select



27 equipment available

BENGKEL CNC

M108

CNC workshop for computer-controlled machining and manufacturing practice.

MANUFACTURING ENGINEERING

27 / 29 equipment available

Select

5. Booking Request Page

[← Back to booking](#)

Complete Booking

Review all selected equipment before submitting one request.

Booking Instructions

- Application shall be made at least 3 working days before usage.
- All information provided in this form must be TRUE upon submission.
- If the equipment requested from applicants are to be brought outside the laboratory / workshop, an application letter endorsed by Supervisor / Project Leader / Lecturer needs to be submitted to the Director of Administration of Deputy Vice-Chancellor (Research & Innovation), UTMKL.
- The office has the right to reject any activity from the applicant if the activities are suspected to have high risks to the staff / environment and/or can cause damage to the instrument.
- For further inquiries on the availability of laboratory and equipment, kindly contact the staff of the respective laboratory.



No equipment selected
Add equipment from the booking page to build your request.

[Browse Labs](#)

Request Details

Estimated total
RM 0
Final price decision is based on PPMU review.

USER DETAILS

Username
zhaowei READ ONLY

Role
student READ ONLY

Email
zhaowei@graduate.utm.my READ ONLY

User ID *

Faculty *

Contact Number *

Study Level *

Select user status ▼

LECTURER DETAILS

Lecturer Name *

Lecturer Email *

Lecturer Contact *

Faculty *

ID Number *

VOT Number *

PIC DETAILS

PIC Token *

6-character code

REQUEST DETAILS

Additional Request Details

[Submit Booking Request](#)

6.Booking Record Page

Booking Records

View and manage submitted booking requests.

Filter by Status
All Statuses



No bookings found
Your submitted booking requests will appear here.

7.Generate Token Page

Token Generation

Generate and assign booking tokens for students or staff.

PIC workflow
1. Search for the user by university email. 2. Generate a 6-character token. 3. Assign the token to that user. 4. The user enters it during lab or equipment booking.

User email

User found successfully.

Find user

User details

Username

Role

Email

Generate code

Generated code

Code: KQUHT4
Expires at: 2026/6/29 23:59:59
Username:
Role: student

Assign

8.View Assigned Token Page

Assigned Tokens

Expire all tokens

Tip: Active status is clickable. Click an active badge to manually expire that token.

ID	Generation Time	Expiry	Token	User Email	Status
36	2026/6/19 11:51:33	2026/6/26 23:59:59	8SFMRP		active
32	2026/6/18 15:46:25	2026/6/25 23:59:59	UH4HR9		expired
30	2026/6/15 18:26:22	2026/6/22 23:59:59	JETE65		expired
28	2026/6/13 21:55:48	2026/6/20 23:59:59	XE73QR		expired
27	2026/6/12 14:28:45	2026/6/19 23:59:59	TRT483		expired

9.Unit Leader Review Page

Unit Leader Recommendation Dashboard

Review booking requests before they move to PPMU final decision.

How to review requests

Pending requests appear first. Use search and filters to narrow the list, open View Details to inspect the booking, then submit the unit leader recommendation.

1

Total Requests Received

0

Pending Review

0

Recommended

0

Rejected

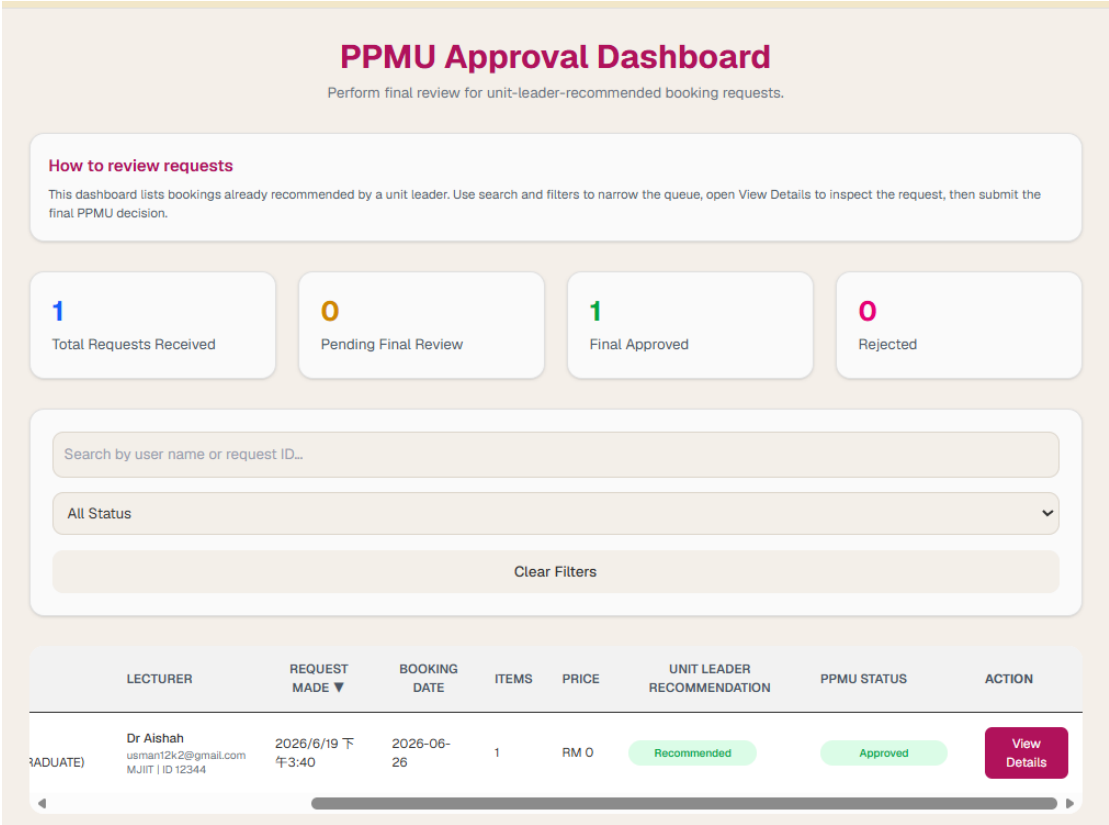
Search by user name or request ID...

All Status

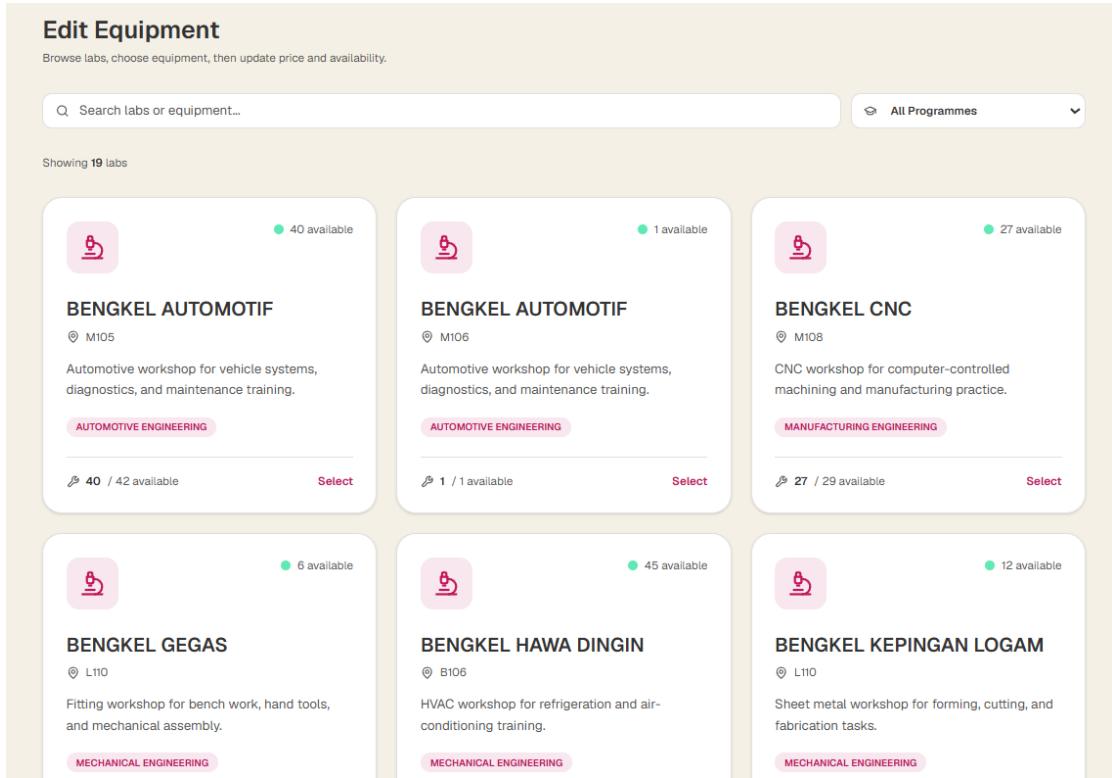
Clear Filters

IDY LEVEL	LECTURER	REQUEST MADE	BOOKING DATE	ITEMS	PRICE	UNIT LEADER RECOMMENDATION	ACTION
IDENT(UNDERGRADUATE)	Dr Aishah usman12k2@gmail.com MUJIT ID 12344	2026/6/19 下午 3:40	2026-06-26	2	RM 0	Partially Reviewed	View Details

10.PPMU Review Page



12. Management Equipment Page



5.0 SOFTWARE DESIGN DESCRIPTION

5.1 System Architecture Backend

- Next.js API Routes are used to handle backend logic.
- Supabase PostgreSQL is used for data storage and retrieval.
- API endpoints are used for authentication, booking requests, PIC token management, Unit Leader review, PPMU review, equipment management, PDF generation, and email notifications.

5.2 Impacts of the Backend Technology Change

- Improved Efficiency:
Booking requests can be submitted, reviewed, approved, rejected, and tracked through one system instead of separate manual communication.
- Reduced Booking Conflicts:
The system checks class timetables, pending requests, approved bookings, and equipment status before allowing a request to be submitted.
- Increased Transparency:
Users can view their booking records, request status, reviewer remarks, and final decision directly in the system.
- Better Role Control:
Student/Staff, PIC, Unit Leader, and PPMU users are given different access based on their responsibilities.
- More Reliable Records:
Booking data, token records, approval decisions, pricing, and generated documents are stored and managed consistently.

5.3 System Architecture Frontend

- **Next.js and React:**
The frontend is developed using Next.js and React to build interactive web pages.
- **Role-Based Interface:**
Different pages and navigation options are shown based on the user role, including Student/Staff, PIC, Unit Leader, and PPMU.
- **Styling and Responsiveness:**
The interface uses CSS/Tailwind-style class names to manage layout, visual design, and responsive display.

5.4 Frontend and Backend Modules

5.4.1 User Authentication Module

This module manages user registration, login, password reset, profile information, and role-based navigation. It ensures that users can only access the features allowed for their role.

5.4.2 Lab and Equipment Browsing Module

This module allows users to browse labs and equipment, search or filter resources, view equipment status, price, staff details, and schedule availability.

5.4.3 Booking Request Module

This module allows Student/Staff users to create booking requests. It supports multiple equipment items in one request, requester and lecturer details, VOT number, study level, PIC token, estimated price calculation, and booking rule validation.

5.4.4 PIC Token Module

This module allows PIC users to generate, assign, view, sort, and expire booking tokens. The token is used to authorize users before they submit booking requests.

5.4.5 Unit Leader Review Module

This module allows Unit Leaders to review submitted booking requests. They can recommend or reject each equipment item and provide remarks before the request is forwarded to PPMU.

5.4.6 PPMU Review Module

This module allows PPMU users to perform final approval. PPMU can approve or reject items, edit final prices, add remarks, and confirm the final request status.

5.4.7 Equipment Management Module

This module allows PPMU users to update equipment price and availability status. Equipment marked as under maintenance or unavailable cannot be booked by users.

5.4.9 Document and Notification Module

This module generates booking receipt and quotation PDFs. It also sends email notifications for booking submission, token usage, Unit Leader recommendation, rejection, final approval, and quotation availability.

5.5 Data Flow

In this web application, the frontend communicates with the backend API whenever the user performs an action. These actions include logging in, browsing equipment, submitting booking requests, generating tokens, reviewing requests, updating equipment, and downloading documents.

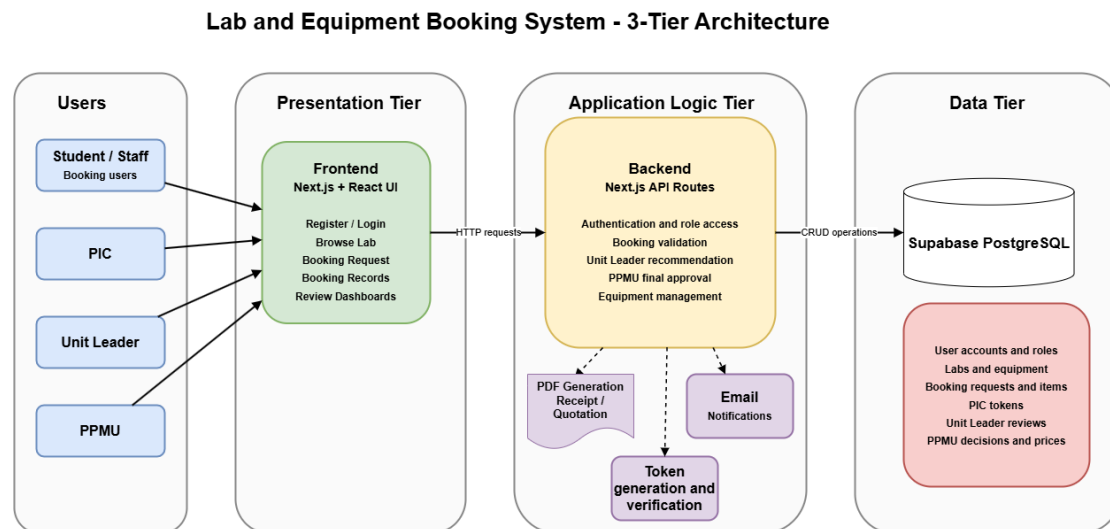
The process begins from the frontend, where user actions trigger requests to the backend API. For example, when a Student/Staff user submits a booking request, the frontend sends the selected equipment, booking date, time, requester details, lecturer details, VOT number, study level, and PIC token to the backend.

After receiving the request, the backend validates the data and communicates with the Supabase PostgreSQL database. It checks whether the PIC token is valid, whether the selected equipment is available, and whether the booking clashes with class timetables, pending requests, or approved bookings. If the request is valid, the backend stores the booking data in the database.

The booking then moves through the approval workflow. Unit Leader reviews the request and gives a recommendation. If recommended, the request is forwarded to PPMU for final review. PPMU makes the final approval or rejection decision, updates final pricing if needed, and confirms the final request status.

The system also updates booking records, generates PDF documents when required, and sends email notifications to related users. This data flow ensures that booking activities are processed clearly, consistently, and efficiently.

5.6 System Architecture Diagram



5.7 Advantages of 3-Tier Architecture

- **Clear separation of responsibilities:**
The system is divided into Presentation Tier, Application Logic Tier, and Data Tier, making each part responsible for a specific function.
- **Easier maintenance:**
Developers can update the user interface, backend logic, or database structure separately without affecting the whole system.
- **Reusable business logic:**
Booking validation, PIC token verification, Unit Leader review, and PPMU approval logic can be reused by different frontend pages through API routes.
- **Better data consistency:**
The backend controls how data is created, updated, and retrieved, helping maintain accurate booking records, token records, and approval status.
- **Easier integration:**
Supporting services such as PDF generation and email notifications can be connected through the backend without changing the frontend or database design.
- **Supports role-based workflow:**
The architecture fits the system because Student/Staff, PIC, Unit Leader, and PPMU all use different interfaces while sharing the same backend workflow and database.

6.0 DATABASE DESIGN DESCRIPTION

6.1 Database Design

user table

Field Name	Data Type	Description
id	int8	Unique ID for each user.
created_at	timestamptz	Date and time when the user account was created.
username	varchar	Username of the user.
email	varchar	Email address of the user.
role	varchar	User role such as student, staff, pic, unit_leader, or ppmu.
verified	bool	Shows whether the account has been verified.
verification_expiry	timestamptz	Expiry date and time for account verification.

pic_tokens table

Field Name	Data Type	Description
id	int8	Unique ID for each PIC token record.
created_at	timestamptz	Date and time when the token was created.
expires_at	timestamptz	Date and time when the token expires.
assigned_to	int8	User ID of the requester assigned to the token.
assigned_by	int8	User ID of the PIC who generated the token.
token	text	Generated PIC token used for booking authorization.
manual_expire	bool	Shows whether the token has been manually expired.

lab table

Field Name	Data Type	Description
id	text	Unique ID for each lab.
name	text	Name of the lab.
description	text	Description of the lab.
location	text	Location of the lab.
price_per_hour	int4	Hourly booking price for the lab.
course	text	Course or programme related to the lab.
created_at	timestamp	Date and time when the lab record was created.
staff_name	text	Name of the staff in charge of the lab.

staff_email	text	Email address of the staff in charge.
staff_contact	text	Contact number of the staff in charge.
status	text	Current lab status, such as available or unavailable.

equipment table

Field Name	Data Type	Description
id	text	Unique ID for each equipment.
name	text	Name of the equipment.
description	text	Description of the equipment.
location	text	Location of the equipment.
status	text	Current equipment status, such as available, maintenance, or unavailable.
price_per_hour	integer	Hourly booking price for the equipment.
course	text	Course or programme related to the equipment.
created_at	timestamp	Date and time when the equipment record was created.
quantity	integer	Quantity of the equipment available.
lab_id	text	ID of the lab linked to the equipment.
staff_name	text	Name of the staff in charge of the equipment.
staff_email	text	Email address of the staff in charge.
staff_contact	text	Contact number of the staff in charge.

equipment_bookings table

Field Name	Data Type	Description
id	bigint	Unique ID for each equipment booking item.
booking_id	bigint	ID of the main booking request linked to this equipment item.
equipment_id	text	ID of the equipment being booked.
lab_id	text	ID of the lab linked to the equipment.
start_time	time	Booking start time.
end_time	time	Booking end time.
booking_reason	text	Purpose or reason for booking the equipment.
status	text	Current status of the equipment booking item, such as pending, under review, approved, rejected, or cancelled.
total_price	numeric	Estimated total price for the equipment booking item.
created_at	timestamptz	Date and time when the equipment booking item was created.
start_date	date	Booking start date.
end_date	date	Booking end date.
new_total_price	bigint	Updated or final total price after review.

bookings table

Field Name	Data Type	Description
id	bigint	Unique ID for each booking request.
user_id	bigint	ID of the user who created the booking request.
booking_date	date	Main booking date of the request.
requester_contact	text	Contact number of the requester.
requester_faculty	text	Faculty of the requester.
requester_identifier	text	Matric ID or staff ID of the requester.
vot_number	text	VOT number used for the booking request.
final_total_price	numeric	Final total price of the booking request.
request_details	text	Additional details or notes for the request.
token	text	PIC token used for the booking request.
overall_status	text	Overall status of the booking request, such as pending, under review, approved, rejected, or cancelled.
created_at	timestampz	Date and time when the booking request was created.
updated_at	timestampz	Date and time when the booking request was last updated.
study_level	text	Study level of the requester.
lect_name	text	Lecturer name.
lect_email	text	Lecturer email address.
lect_contact	text	Lecturer contact number.
pic_email	text	Email address of the PIC.
pic_name	text	Name of the PIC.
lect_faculty	text	Lecturer faculty.
lect_id	text	Lecturer ID.

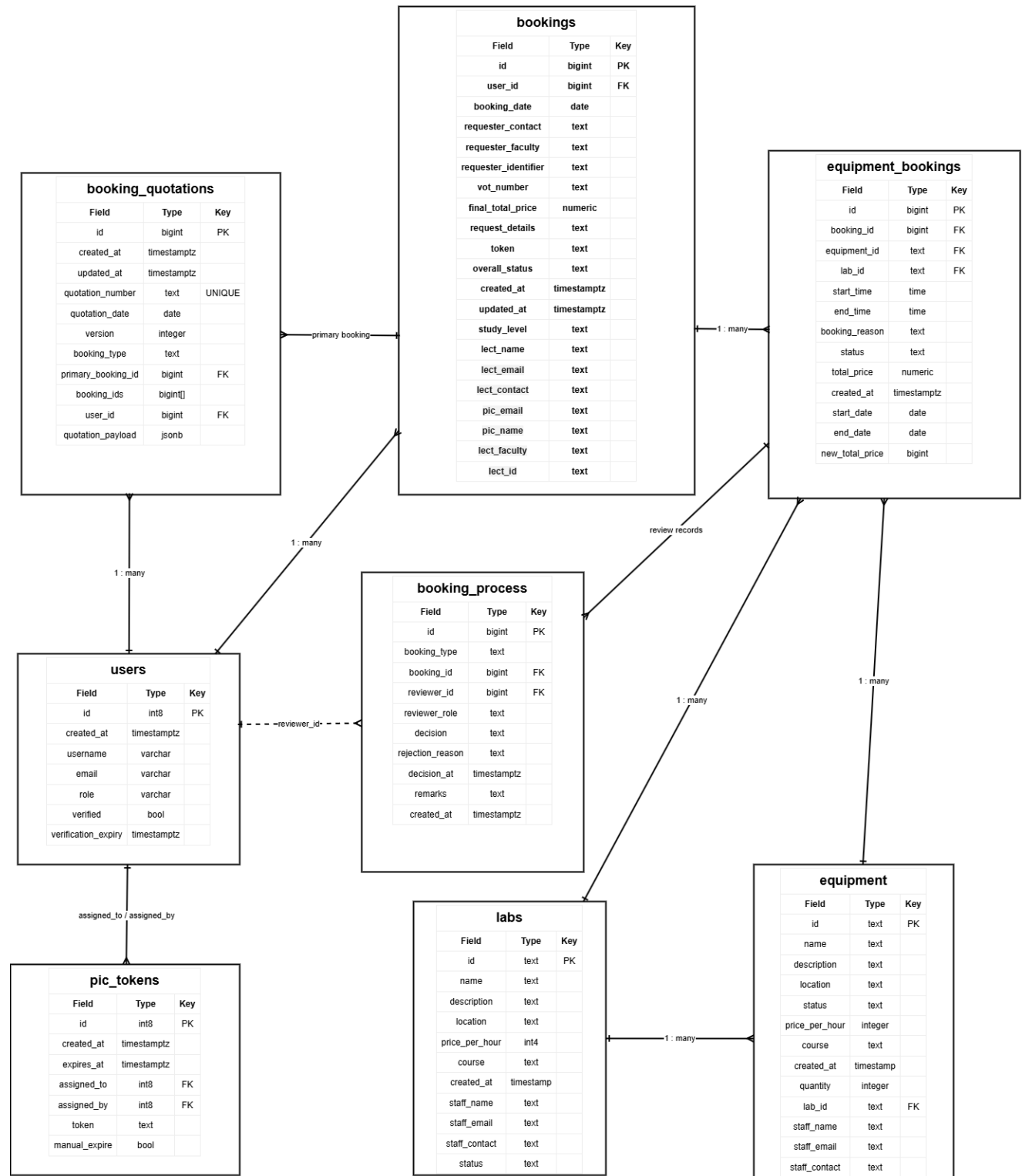
booking_process table

Field Name	Data Type	Description
id	bigint	Unique ID for each booking review process record.
booking_type	text	Type of booking being reviewed, either lab or equipment.
booking_id	bigint	ID of the related booking item or booking request.
reviewer_id	bigint	ID of the reviewer who made the decision.
reviewer_role	text	Role of the reviewer, either unit_leader or ppmu.
decision	text	Review decision, either approved or rejected.
rejection_reason	text	Reason for rejection if the request is rejected.
decision_at	timestampz	Date and time when the decision was made.
remarks	text	Additional remarks from the reviewer.
created_at	timestampz	Date and time when the review record was created.

booking_quotations table

Field Name	Data Type	Description
id	bigint	Unique ID for each quotation record.
created_at	timestampz	Date and time when the quotation was created.
updated_at	timestampz	Date and time when the quotation was last updated.
quotation_number	text	Unique quotation reference number.
quotation_date	date	Date when the quotation was generated.
version	integer	Version number of the quotation.
booking_type	text	Type of booking, either lab or equipment.
primary_booking_id	bigint	Main booking ID linked to the quotation.
booking_ids	bigint[]	List of booking IDs included in the quotation.
user_id	bigint	ID of the user linked to the quotation.
quotation_payload	jsonb	Quotation details stored in JSON format, including requester, booking, item, and pricing information.

6.2 Entity Relationship Diagram



7.0 CODES

The project source code and supporting technical documents are stored in the GitHub repository. The repository includes the frontend source code, backend API routes, Supabase PostgreSQL database schema, PDF generation functions, email notification functions, README file, and UAT testing documents.

The main implementation includes:

- Frontend source code developed using Next.js and React
- Backend API routes developed using Next.js API routes
- Supabase PostgreSQL database schema
- Authentication, booking, token, review, approval, and equipment management modules
- Receipt and quotation PDF generation
- Email notification implementation
- README file for installation and directory explanation
- UAT testing documents

Code Repository: <https://github.com/KaifHalak/AD-Project>

User Manual: <https://github.com/zhaovee/Project-Documentation-Repository/blob/main/User%20Manual.pdf>

UAT: <https://github.com/zhaovee/Project-Documentation-Repository/blob/main/UAT.pdf>

8.0 TEST RESULTS BASED ON USER STORIES

US001 Student/Staff Part

Test Results: Date: 19/June/2026

Tested by: Usman

Tested Application: The feature to log in as a student or staff

Project Name: Lab and Equipment Booking System					
Criteria	Responsible Party	Priority	Test Date	Response	Comments
User can register an account using valid university email.		High	19/June/2026, 3:00 PM	Pass	
System rejects registration with invalid or duplicate email.		High	19/June/2026, 3:00 PM	Pass	
User can log in with valid email and password.		High	19/June/2026, 3:00 PM	Pass	
System rejects invalid login credentials.		High	19/June/2026, 3:00 PM	Pass	
User can reset or update password successfully.		Medium	19/June/2026, 3:00 PM	Pass	
User can browse labs and equipment.		High	19/June/2026, 3:00 PM	Pass	
User can search or filter labs and equipment.		Medium	19/June/2026, 3:00 PM	Pass	
User can view lab/equipment details, price, status, and schedule.		High	19/June/2026, 3:00 PM	Pass	
System blocks unavailable or maintenance equipment from booking.		High	19/June/2026, 3:00 PM	Pass	
User can add one or more equipment items into a booking request.		High	19/June/2026, 3:00 PM	Pass	
User can submit booking request with complete required details.		High	19/June/2026, 3:00 PM	Pass	
System prevents submission with missing details or invalid PIC token.		High	19/June/2026, 3:00 PM	Pass	
System prevents booking conflicts with timetable or existing bookings.		High	19/June/2026, 3:00 PM	Pass	
User can view submitted booking records and status.		High	19/June/2026, 3:00 PM	Pass	
User can cancel eligible pending booking request.		Medium	19/June/2026, 3:00 PM	Pass	
User can download booking receipt or quotation PDF.		Medium	19/June/2026, 3:00 PM	Pass	
User receives booking submission, approval, or rejection notification.		Medium	19/June/2026, 3:00 PM	Pass	

US002 PIC Part

Test Results: Date: 19/June/2026

Tested by: ABDULRAHMAN SIAD

Tested Application: The feature to log in as a PIC

Project Name:Lab and Equipment Booking System					
Criteria	Responsible Party	Priority	Test Date	Response	Comments
PIC can log in and access PIC features.		High	19/June/2026, 3:00 PM	Pass	
PIC can search for a registered user by email.		High	19/June/2026, 3:00 PM	Pass	
PIC can generate a 6-character booking token.		High	19/June/2026, 3:00 PM	Pass	
PIC can assign generated token to the selected user.		High	19/June/2026, 3:00 PM	Pass	
System prevents non-PIC users from accessing token generation.		High	19/June/2026, 3:00 PM	Pass	
PIC can view assigned token records.		Medium	19/June/2026, 3:00 PM	Pass	
PIC can view token details such as token code, user, created date, and expiry date.		Medium	19/June/2026, 3:00 PM	Pass	
PIC can manually expire an active token.		Medium	19/June/2026, 3:00 PM	Pass	
Expired token cannot be used for booking request.		High	19/June/2026, 3:00 PM	Pass	
PIC receives notification when assigned token is used.		Medium	19/June/2026, 3:00 PM	Pass	
PIC receives notification when related request is approved or rejected.		Medium	19/June/2026, 3:00 PM	Pass	

US002 Unit Leader Part

Test Results: Date: 19/June/2026

Tested by:Zhao Wei

Tested Application: The feature to log in as a unit leader

Project Name:Lab and Equipment Booking System					
Criteria	Responsible Party	Priority	Test Date	Response	Comments
Unit Leader can log in and access Unit Leader dashboard.		High	19/June/2026, 3:00 PM	Pass	
Dashboard displays pending requests and request statistics.		High	19/June/2026, 3:00 PM	Pass	
Unit Leader can search, filter, and sort booking requests.		Medium	19/June/2026, 3:00 PM	Pass	
Unit Leader can open request detail page.		High	19/June/2026, 3:00 PM	Pass	
Unit Leader can view requester, lecturer, PIC, equipment, and staff details.		High	19/June/2026, 3:00 PM	Pass	
Unit Leader can recommend individual equipment item.		High	19/June/2026, 3:00 PM	Pass	
Unit Leader can reject individual equipment item with remarks.		High	19/June/2026, 3:00 PM	Pass	
Unit Leader can recommend all or reject all reviewable items.		Medium	19/June/2026, 3:00 PM	Pass	
System prevents duplicate recommendation or rejection.		High	19/June/2026, 3:00 PM	Pass	
Recommended items are forwarded to PPMU review.		High	19/June/2026, 3:00 PM	Pass	
System prevents non-Unit Leader users from accessing Unit Leader features.		High	19/June/2026, 3:00 PM	Pass	
Unit Leader receives notification after PPMU final decision.		Medium	19/June/2026, 3:00 PM	Pass	

US002 Unit Leader Part

Test Results: Date: 19/June/2026

Tested by: Usman

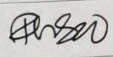
Tested Application: The feature to log in as a PPMU

Project Name: Lab and Equipment Booking System					
Criteria	Responsible Party	Priority	Test Date	Response	Comments
PPMU can log in and access PPMU dashboard.		High	19/June/2026, 3:00 PM	Pass	
Dashboard displays requests ready for final review and statistics.		High	19/June/2026, 3:00 PM	Pass	
PPMU can search, filter, and sort requests.		Medium	19/June/2026, 3:00 PM	Pass	
PPMU can open request detail page.		High	19/June/2026, 3:00 PM	Pass	
PPMU can view full requester, lecturer, PIC, equipment, and staff details.		High	19/June/2026, 3:00 PM	Pass	
PPMU can approve recommended equipment item.		High	19/June/2026, 3:00 PM	Pass	
PPMU can reject recommended equipment item with remarks.		High	19/June/2026, 3:00 PM	Pass	
PPMU can edit final price before approval.		High	19/June/2026, 3:00 PM	Pass	
System blocks PPMU review for items not recommended by Unit Leader.		High	19/June/2026, 3:00 PM	Pass	
PPMU can update equipment price.		Medium	19/June/2026, 3:00 PM	Pass	
PPMU can set equipment status to available, maintenance, or unavailable.		Medium	19/June/2026, 3:00 PM	Pass	
Updated equipment status is reflected in the booking page.		High	19/June/2026, 3:00 PM	Pass	
System prevents non-PPMU users from accessing PPMU features.		High	19/June/2026, 3:00 PM	Pass	
PPMU final decision triggers notifications to related users.		Medium	19/June/2026, 3:00 PM	Pass	
System generates quotation PDF after final approval.		Medium	19/June/2026, 3:00 PM	Pass	

Stakeholder Signature:

7. User Acceptance Sign-Off

The undersigned confirm that the system has been tested according to the agreed User Stories and Acceptance Criteria.

Name	Role	Signature	Date
ROHANA BINI SAMP	End User		19/6/2026
	Project Supervisor		
	Development Team Representative		

Final Decision

☐ Accepted

☒ Accepted with Minor Issues

☐ Rejected

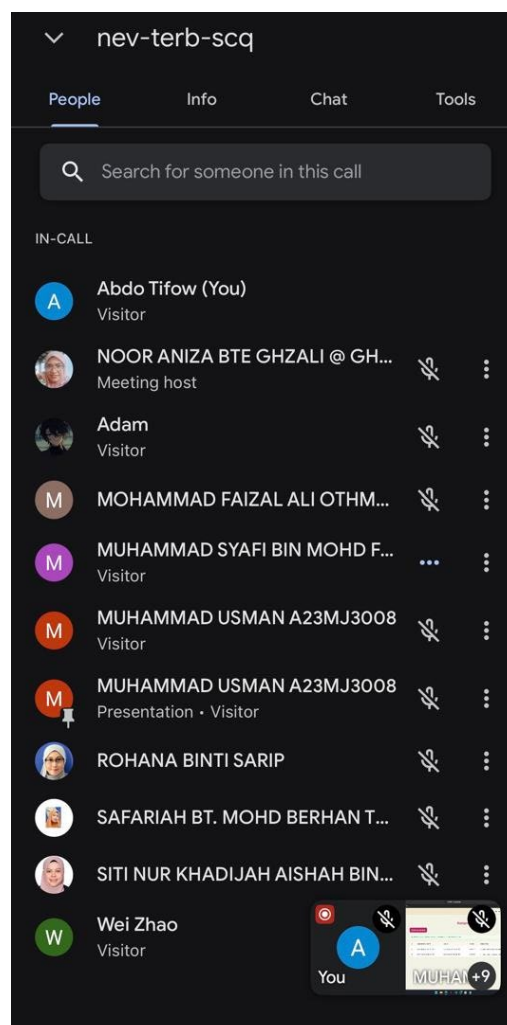
Remark:

APPENDIX

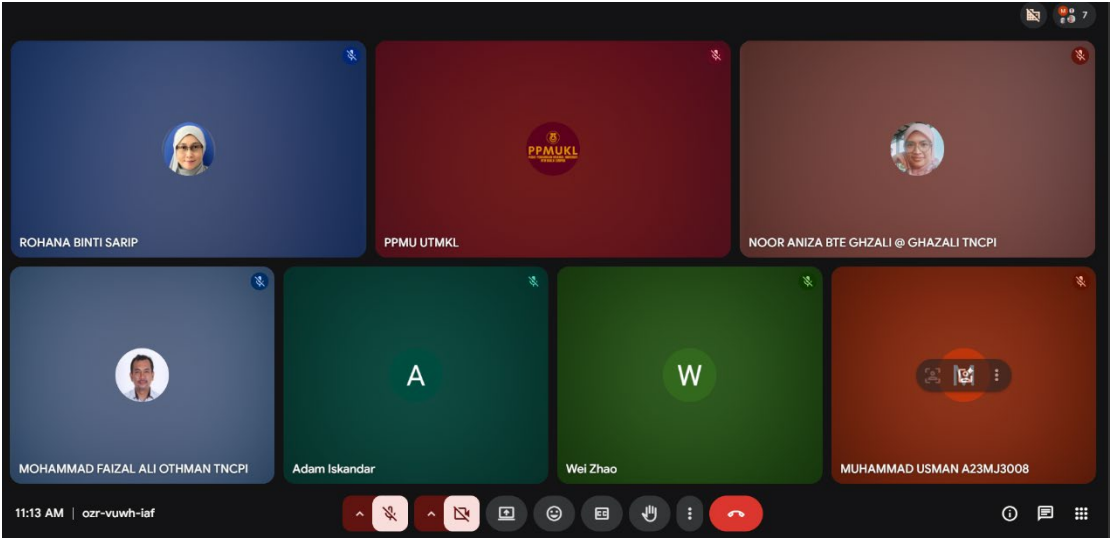
First offline meeting for sprint 1



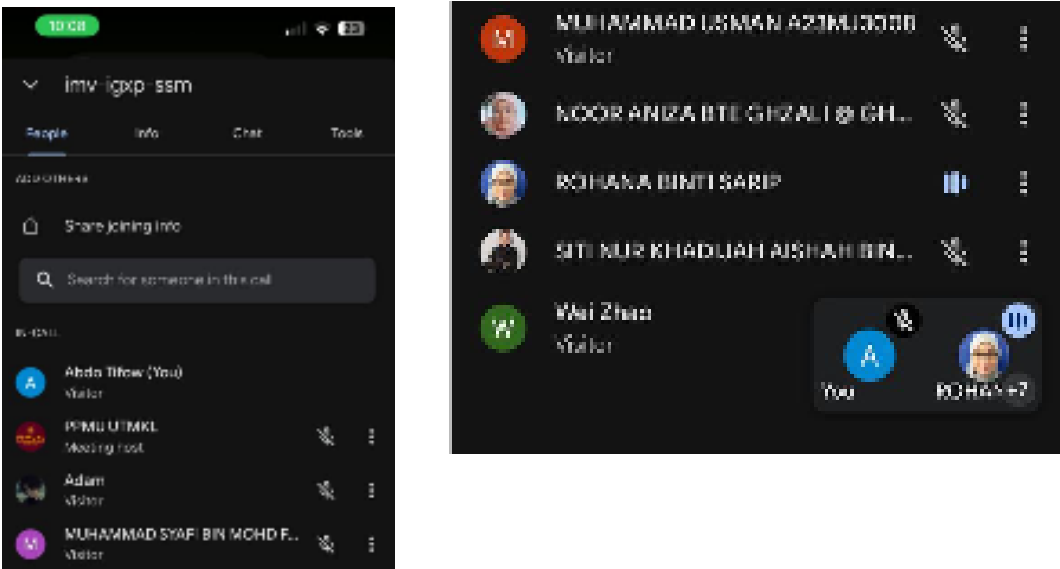
Second online meeting for sprint 2



Third online meeting for sprint 3



Fourth online meeting for sprint 4



Physical Testing Meeting



Task assignment example (photo)

Laboratory Booking System

SummaryUserBookingInventoryCalendarBookingCodeReportsDocsSprintsDevelopment

Search tasks...Filter

SPRINT Sprint 3 1 Dec - 21 Dec (Approval Workflow & Billing Information System)

S3-TASK-01	Setup Sprint 3 and Define User Stories	DONE ✓	AS	...
S3-TASK-02	Design Approval Workflow (Unit Leader & PPMU)	DONE ✓	AS	...
S3-TASK-03	Implement Unit Leader Approval	DONE ✓	AS	...
S3-TASK-04	Implement PPMU Approval	DONE ✓	AS	...
S3-TASK-05	Display Approval Status & Rejection Information	DONE ✓	AS	...
S3-TASK-06	Implement Billing Information (Grant & VOT Number)	DONE ✓	AS	...
S3-TASK-07	Role-Based Access for Unit Leader and PPMU	DONE ✓	AS	...
S3-TASK-08	Notification for Approval / Rejection	DONE ✓	AS	...
S3-TASK-09	Testing and Bug Fixing for Approval Workflow	DONE ✓	AS	...
S3-TASK-10	Documentation and Sprint 3 Report	DONE ✓	AS	...

+ Create